Topological Collapse and Asymmetric Divergence of Carmichael Pseudoprimes on the $6N$ Skeleton*

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Abstract

On the $6N$ skeleton every integer coprime to 6 sits on a wing: the right wing $6N+1$ ($\equiv 1 \bmod 6$) or the left wing $6N-1$ ($\equiv 5 \bmod 6$). We read the Carmichael numbers through this geometry. Korselt's criterion forces a sharp, elementary structure that we state and verify: *(i)* a one-way semipermeable membrane—every Carmichael number on the left wing has *all* of its prime factors on the left wing (and is not divisible by 3); *(ii)* a parity law—a Carmichael number coprime to 6 lies on the left wing iff it has an odd number of left-wing prime factors; and *(iii)* a unified barrier—whenever $3 \nmid (n-1)$ no right-wing prime can divide n, which fuses the left wing with the off-skeleton residue-3 numbers (a residue-3 Carmichael number is exactly a 3 glued to a pure-left core). Enumerating all 1547 Carmichael numbers up to 10^{10}, all three laws hold with *zero* violations. The geometric consequence is a violent asymmetry: the right-to-left ratio $R(X) = \#\{\text{right}\}/\#\{\text{left}\}$ climbs monotonically from 5.0 at $X = 10^4$ to 88.8 at $X = 10^{10}$, a collapse of the left-wing population across seven decades. The same monochromatic suppression acts *inside* the right wing: the pure-right family is overtaken by the mixed family, the internal ratio $M(X) = \#\{\text{pure-right}\}/\#\{\text{mixed}\}$ crossing unity between 10^7 and 10^8. We trace the mechanism to a single structural fact—single-colour cores require all factors in one wing, while the mean number of prime factors $\bar{k}(X)$ grows ($3.00 \to 4.23$)—but we are explicit that the two single-colour basins are *not* symmetric: pure-right decays about 2.6 times more slowly than the left wing in the log-exponent, because right-wing primes have smooth ($6 \mid p-1$) and left-wing primes rough ($3 \nmid p-1$) predecessors. We claim no unified thermodynamic law: there is no single decay exponent, and the divergence *rate* is itself not identifiable—a power of $\log X$, a small power of X, and a stretched exponential fit equally well, and Poisson sparsity plus the deep pre-asymptotic character of Carmichael statistics put the asymptotic law beyond any reachable window. What is unified is *algebraic* (the single barrier of law *(iii)*), not thermodynamic; the structural laws and the asymmetric-collapse phenomenon are firm, the asymptotic divergence law is, by construction, beyond reach.

1 Introduction

A Carmichael number is a composite n for which $a^n \equiv a \pmod{n}$ for every integer a; by Korselt's criterion [1] these are exactly the squarefree n with at least three prime factors such that $p-1 \mid n-1$ for every prime $p \mid n$. They are infinite in number [3], but their finer distribution—the growth of $C(x) = \#\{\text{Carmichael} \le x\}$ and the internal correlations among their prime factors—remains, in Erdős's heuristic [4] and in Pinch's tabulations [5], only partly understood.

*Code and measured data: https://github.com/Ruqing1963/6N-carmichael-collapse (Part XXXVI).

The present series reads arithmetic objects through the $6N$ skeleton, on which every prime > 3 lies on a fault line $6N \pm 1$. Here we ask what Korselt's criterion looks like in that geometry. The answer is an elementary but unusually rigid structure, and a measured macroscopic consequence. We make no claim that the structural facts are new as mathematics—they follow in a few lines from Korselt—only that their statement in the wing coordinates is clean, and that the resulting population asymmetry, and its honest non-identifiability, are worth recording.

2 The structural laws

Write a prime $p > 3$ as *right* if $p \equiv 1 \pmod 6$ and *left* if $p \equiv 5 \pmod 6$. On the units $(\mathbb{Z}/6)^\times = \{1, 5\} \cong \{\pm 1\}$, right primes carry the residue $+1$ and left primes the residue -1.

Proposition 1 (The semipermeable membrane)**.** *Let n be a Carmichael number with $n \equiv 5 \pmod 6$ (the left wing). Then $3 \nmid n$ and every prime factor of n satisfies $p \equiv 5 \pmod 6$. No right-wing prime, and no factor of 3, can occur.*

Proof. $n \equiv 5 \pmod 6$ gives n odd and $3 \nmid n$, so n is coprime to 6 and every prime factor is $\equiv 1$ or $\equiv 5 \pmod 6$. Suppose $p \mid n$ with $p \equiv 1 \pmod 6$. Then $6 \mid p - 1$. By Korselt $p - 1 \mid n - 1$, hence $6 \mid n - 1$. But $n \equiv 5 \pmod 6$ gives $n - 1 \equiv 4 \pmod 6$, so $6 \nmid n - 1$—a contradiction. Thus no factor is $\equiv 1 \pmod 6$, and all factors are $\equiv 5 \pmod 6$. $\square$

Proposition 2 (The parity law)**.** *Let n be a Carmichael number coprime to 6, with ℓ prime factors $\equiv 5 \pmod 6$ (counted with the multiplicity 1, as n is squarefree). Then $n \equiv (-1)^\ell \pmod 6$. Equivalently: n is on the right wing iff ℓ is even, and on the left wing iff ℓ is odd.*

Proof. Modulo 6 the residue of a product of units is the product of residues; right factors contribute $+1$ and left factors -1, so $n \equiv (+1)^{\#\text{right}}(-1)^\ell \equiv (-1)^\ell \pmod 6$. $\square$

Corollary 1. *A left-wing Carmichael number is a product of an* odd *number (≥ 3) of distinct primes, all $\equiv 5 \pmod 6$. The left wing is the single monochromatic, odd-length family; the right wing collects everything else—the all-right products, and all mixed products with an even number of left factors.*

The membrane of Proposition 1 is not special to the left wing. Its true cause is the divisibility of $n - 1$ by 3, and stating it that way unifies the left wing with the third class—the Carmichael numbers divisible by 3, which leave the skeleton at residue $3 \pmod 6$ (e.g. $561 = 3 \cdot 11 \cdot 17$).

Proposition 3 (The unified barrier)**.** *Let n be a Carmichael number with $3 \nmid (n - 1)$. Then n has no prime factor $\equiv 1 \pmod 6$. Among the residues coprime to 3 this is exactly $n \equiv 5 \pmod 6$ (left wing) and $n \equiv 3 \pmod 6$ ($3 \mid n$); in both cases every prime factor other than a possible single 3 is $\equiv 5 \pmod 6$. Consequently a residue-3 Carmichael number is precisely 3 times a product of distinct primes all $\equiv 5 \pmod 6$—a 3 glued to a pure-left core—while a left-wing Carmichael number is a pure-left core alone.*

Proof. Suppose $p \mid n$ with $p \equiv 1 \pmod 6$, so $3 \mid p - 1$. By Korselt $p - 1 \mid n - 1$, hence $3 \mid n - 1$, against the hypothesis. So no factor is $\equiv 1 \pmod 6$. Now $3 \mid n - 1$ fails exactly when $n \not\equiv 1 \pmod 3$, i.e. $n \equiv 2 \pmod 3$ ($\Leftrightarrow n \equiv 5 \pmod 6$, the left wing) or $3 \mid n$ ($\Leftrightarrow n \equiv 3 \pmod 6$). In the first case $3 \nmid n$, so all factors are $\equiv 5 \pmod 6$ (recovering Proposition 1); in the second, $n = 3m$ with m squarefree and coprime to 3, and the argument applied to each prime of m forces every such prime to be $\equiv 5 \pmod 6$. $\square$

Remark 1. The complementary case is $3 \mid n-1$, i.e. $n \equiv 1 \pmod 6$ (the right wing), where $p-1 \equiv 0 \pmod 6$ is permitted and right-wing factors are admitted. So a single criterion governs all three classes: the divisor 3 of $n-1$ is the gate. When it is absent, every right-wing prime is barred and the factor core is forced fully onto the left wing (with, in the residue-3 case, an extra 3); when it is present, the mixed basin opens. The "ghost" residue-3 numbers are therefore not a separate phenomenon but the $3 \mid n$ face of the same barrier.

These structural facts are elementary consequences of Korselt's criterion; we claim only their clean statement and verification, not novelty as theorems. Their force is geometric: whenever $3 \nmid (n-1)$ the right wing is sealed off and the factor core collapses to a single colour, while $3 \mid (n-1)$ opens the mixed basin.

3 The measurement

We enumerated *all* Carmichael numbers up to 10^{10} by the Korselt divisor method (for each ordered prefix of prime factors with product M, the final prime r must satisfy $(r-1) \mid (M-1)$, checked against $p-1 \mid n-1$ for the remaining factors), recovering the known count $C(10^{10}) = 1547$ [5] exactly, with the correct factorisation of each. Across all 1547 numbers, Propositions 1, 2, and 3 hold with **zero violations**: every left-wing Carmichael number is monochromatically left, every right-wing Carmichael number has an even number of left-wing factors, and every one of the 20 residue-3 numbers up to 10^{10} is a 3 glued to a pure-left core (e.g. $561 = 3 \cdot 11 \cdot 17$, $62745 = 3 \cdot 5 \cdot 47 \cdot 89$).

Table 1 reports the cumulative wing census by decade. The right-to-left ratio $R(X) = \#\{n \leq X : n \equiv 1\}/\#\{n \leq X : n \equiv 5\}$ rises monotonically from 5.0 to 88.8. (The residue-3 class, governed by Proposition 3, is small—20 numbers up to 10^{10}—and is excluded from the wing ratio.)

$\log_{10} X$	$C(X)$	right	left	$R(X)$	$\bar{k}(X)$
4	7	5	1	5.00	3.00
5	16	13	1	13.00	3.25
6	43	38	2	19.00	3.49
7	105	98	4	24.50	3.58
8	255	243	6	40.50	3.78
9	646	623	12	51.92	4.00
10	1547	1510	17	88.82	4.23

Table 1: Cumulative wing census of Carmichael numbers. $R(X)$ is the right/left ratio; $\bar{k}(X)$ is the mean number of prime factors. The left-wing population collapses relative to the right as X grows.

4 Mechanism: monochromatic suppression under a growing factor count

By Corollary 1 a left-wing Carmichael number must be *monochromatic*: all of its k prime factors must fall on the left wing, with k odd. For a Carmichael number with k factors, the heuristic chance that all k are left-wing is of order 2^{-k} (the two wings carry primes of nearly equal density). The mean number of prime factors of Carmichael numbers grows with scale—we measure $\bar{k}(X)$ rising from 3.00 to 4.23 across the range (Table 1)—so the monochromatic condition becomes exponentially rarer, and the left wing is squeezed.

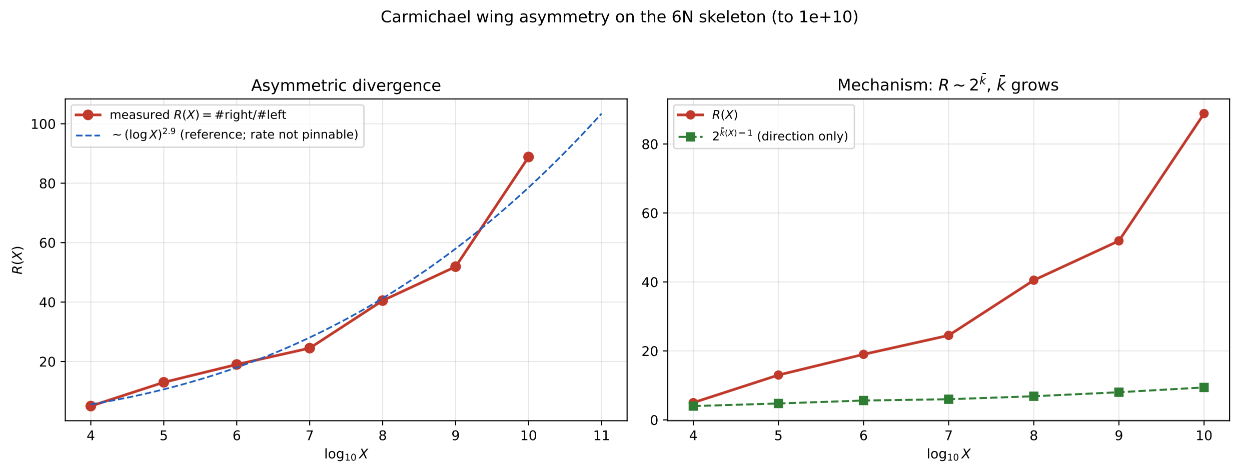

Figure 1: (Left) The measured asymmetric divergence $R(X)$ over seven decades, with a best-fitting power of $\log X$ drawn for reference only—the rate is not identifiable (§6). (Right) The mechanism: $R(X)$ tracks the growth of the mean prime-factor count $\bar{k}(X)$ through the monochromatic suppression $\sim 2^{-\bar{k}}$; the curve $2^{\bar{k}-1}$ captures the direction but undershoots the measured magnitude, because the right wing additionally absorbs the pure-right and even-mixed families.

This is the whole of the picture, and it is only directional. The naive estimate $R \sim 2^{\bar{k}}$ rises far more slowly than the data (Fig. 1, right): $2^{\bar{k}-1}$ runs from 4 to 9.4 while R runs from 5 to 89. The shortfall is structural—the right wing is not merely "not-all-left" but also harvests the entire all-right family and every even-length mixed product—so R outruns $2^{\bar{k}}$. We name the qualitative picture a *topological entropy increase*: a low-entropy, fully ordered (single-colour) basin being absorbed into a high-entropy, mixed basin. We use the phrase only as a mnemonic for the measured monochromatic suppression; it names a counting asymmetry, not a dynamical law, and no entropy functional is claimed.

5 Inside the right basin: pure-right versus mixed

The mechanism above applies to *any* single-colour core, not only the left wing. The all-right products (every factor $\equiv 1 \pmod 6$) are equally monochromatic, and by the same heuristic they too should be squeezed by the mixed products as $\bar{k}$ grows. We test this by splitting the right wing into its pure-right part ($\ell = 0$ left factors) and its mixed part ($\ell \geq 2$, necessarily even), and tracking the internal ratio $M(X) = \#\{\text{pure-right}\}/\#\{\text{mixed}\}$.

Table 2 confirms the prediction in direction. Pure-right leads at small scale ($M \approx 1.5$), crosses the topological transition $M = 1$ between 10^7 and 10^8, and falls to $M = 0.64$ at 10^{10}, where pure-right is only 39% of the right wing. The monochromatic suppression therefore acts on *both* single-colour basins: the left wing collapses outright, and the pure-right family, though it survives on the right, is itself overtaken by the mixed state.

But the two single-colour basins are *not* symmetric, and we are explicit about it. A power-of-log fit gives $M(X) \sim (\log X)^{-1.1}$, whereas the left-wing collapse runs as $R(X) \sim (\log X)^{+2.9}$: pure-right decays about 2.6 times more slowly, in the log-exponent, than the left wing. The cause is arithmetic, not thermodynamic. A right-wing prime has $p - 1 \equiv 0 \pmod 6$, divisible by $6 = 2 \cdot 3$ and hence "smooth"—rich in small divisors, easy to align under the common-divisor requirement $p - 1 \mid n - 1$. A left-wing prime has $p - 1 \equiv 4 \pmod 6$, divisible by 2 but *not* by 3, hence "rougher"

$\log_{10} X$	pure-right	mixed	$M(X)$
4	3	2	1.50
5	8	5	1.60
6	23	15	1.53
7	53	45	1.18
8	114	129	0.88
9	254	369	0.69
10	587	923	0.64

Table 2: The right basin splits into a monochromatic pure-right family and a mixed family. The ratio $M(X)$ crosses unity between 10^7 and 10^8: pure-right leads at small scale and is overtaken by the mixed state thereafter.

and harder to align, and the left wing additionally demands an odd factor count. Pure-right cores are thus easier to assemble and more resistant to collapse than pure-left cores. The single-colour basins decay in the same *direction* but at materially different *rates*.

We therefore do *not* claim a unified thermodynamic law. There is no single exponent governing the decay of monochromatic states: $M(X)$ and $R(X)$ diverge from one another by a factor ~ 2.6 in their log-rates, and (§6) neither rate is asymptotically identifiable. What is unified is *algebraic*, not thermodynamic—the single barrier of Proposition 3.

6 Honest ledger: the divergence rate is unidentifiable

The divergence of $R(X)$ is firm, and its sign and mechanism are explained. Its *rate* is not recoverable, and we state this without hedging.

Fitting the seven points to three candidate laws gives nearly indistinguishable residuals in $\log R$:

$$R \sim (\log X)^{2.9} \text{ (resid 0.12)}, \qquad R \sim X^{0.19} \text{ (resid 0.16)}, \qquad R \sim e^{1.5\sqrt{\log X}} \text{ (resid 0.13)}.$$

No reasonable amount of additional data breaks this degeneracy. Two obstructions compound. First, the left-wing counts are tiny—$1, 1, 2, 4, 6, 12, 17$ across the decades—so the ratios carry Poisson relative errors of order 10–100%; the early points are essentially single observations. Second, Carmichael statistics are deeply pre-asymptotic: the Erdős count carries $\log\log\log / \log\log$ corrections that do not stabilise until astronomically large X, and the factor-count $\bar{k}(X)$ that drives the mechanism creeps upward by less than half a unit per several decades. Extending the census by one or two further decades would roughly double the left-wing count, nowhere near enough to separate a power of $\log X$ from a small power of X. Accordingly we report $R(X)$ as a *measured divergent trend* with a structural cause, and we make *no* claim of an asymptotic divergence law: in any computational window a human can reach, that law is unidentifiable.

7 Conclusion

Read on the $6N$ skeleton, the Carmichael numbers obey three rigid structural laws—a one-way semipermeable membrane (Proposition 1), a parity selection rule (Proposition 2), and a unified barrier (Proposition 3) that fuses the left wing and the residue-3 ghosts under the single criterion $3 \nmid (n-1)$—all verified to 10^{10} with zero violations. The geometric consequence is an asymmetric collapse of the single-colour populations: the right/left ratio $R(X)$ rises from 5.0 to 88.8 across seven decades, and inside the right wing the pure-right family is itself overtaken by the mixed family, $M(X)$ crossing unity between 10^7 and 10^8. Both are driven by the exponential cost of a monochromatic core as the typical prime-factor count grows. We claim only this, and we draw the line sharply. The unification is *algebraic*—the single barrier $3 \nmid (n-1)$, an elementary Korselt consequence, cleanly stated in wing coordinates and confirmed empirically—and *not* thermodynamic: the two single-colour basins collapse in the same direction but at materially different rates ($M \sim (\log X)^{-1.1}$ against $R \sim (\log X)^{+2.9}$, a 2.6-fold gap traced to the smoothness of $p-1$ for right primes versus its roughness for left primes), so no single decay exponent governs them; and the precise asymptotic rate of either is, by Poisson sparsity and pre-asymptotic drift, not identifiable. We assert no unified thermodynamic law and no asymptotic divergence law. The framework says nothing about the infinitude or the true counting function of Carmichael numbers beyond what [3, 4, 5] already give.

Data and code availability. The complete enumeration to 10^{10} (the Korselt divisor method, the wing census, the $M(X)$ split, and the verifiers for all three structural laws) is openly available at `https://github.com/Ruqing1963/6N-carmichael-collapse`.

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